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tó una usabilidad promedio excelente, con variaciones según el contexto de actuación, lo que indica una mayor facilidad de uso por parte de los la necesidad de ajustes para consolidarla como una herramienta eficaz en la gestión de inmunobiológicos especiales.
en salud; Atención primaria de salud; Evaluación de la tecnología biomédica; Vacunación; Esquemas de inmunización.

INTRODUCTION

The Reference Centers for Special Immunobiologicals (CRIEs) were implemented by the Brazilian Ministry of Health in 1993, with the objective of making high-cost vaccines available to people with special clinical conditions, monitoring adverse events following vaccination, and distributing immunobiologicals to other health units[1]. Although Brazil is among the three leading countries in Latin America in vaccinating populations with special needs or clinical situations[2], the centralization of CRIEs has limited access for the population living far from these centers[3]. Expanding the number of units does not fully solve the problem [3]; it is necessary to integrate services[4], especially Primary Health Care (PHC), whose accessibility, knowledge of the reality, and local planning capacity favor the expansion of vaccination coverage[5].

To keep the vaccination schedule up to date in immunosuppressed patients, a study suggests that PHC units strengthen coordination with CRIEs[6]. This need was highlighted in a study of patients undergoing liver transplantation and indicated for special vaccination schedules, which revealed that 82.2% sought vaccination at Basic Health Units (UBSs), while only 13.7% sought vaccination at CRIEs. Furthermore, the health professionals involved in the care of these patients themselves frequently recommended vaccination at UBSs[7].

However, a study carried out in Fortaleza/Ceará, showed that nursing professionals in PHC vaccination rooms do not make requests for special immunobiologicals due to lack of knowledge of the request flow[8]. This highlights the need for efficient planning for the management of vaccination actions involving the acquisition, storage, distribution and administration of immunobiologicals, in addition to monitoring the flow of information across the various levels of management and execution [1].

The CRIE of Hospital de Clínicas (HC) of Universidade Estadual de Campinas (UNICAMP) adopts, as a strategy to expand population access, the distribution of special immunobiologicals on demand to health units in several municipalities, including the UBSs of Campinas, São Paulo, Brazil. However, there is no standardization regarding the request for these immunobiologicals or their monitoring in PHC.

As a strategic resource, the adoption of digital technologies has proven to be fundamental for the generation of area [9], in addition to contributing to the optimization of the organization of services[10]. Among technologies, software stands out as essential computer programs for the dissemination and transformation of information[11]. They must be developed and evaluated with a focus on their usability, which determines the ease of user interaction with the system[12]. Research that evaluated applications and software aimed at the practice of health professionals indicates that technologies with high usability provide intuitive navigation and have high potential for application in daily practice[13–14].

In this context, the creation of the ConectAPS-CRIE digital platform aims to fill an operational gap in the integration between services. This software is designed to assist in the management of special immunobiologicals in PHC. Validated for its functional performance and technical quality, it is a strategy to improve the organization of special immunobiological requests and the vaccination coverage of its specific target audience[15]. Therefore, evaluating the usability of ConectAPS-CRIE in practice, based on a pilot application, is essential for the tool to be not only technically functional, but also intuitive, effective and suited to the needs of healthcare professionals, encouraging their adherence and facilitating their use in routine services.

In view of the above, the present study aimed to evaluate the usability of the ConectAPS-CRIE platform in the management of special immunobiologicals in primary health care by nursing professionals.

METHOD

This is a study to evaluate the usability of the ConectAPS-CRIE biomedical technology, developed with the expectation that by creating a computer system, the developer will investigate the ease with which users can perform specific tasks when interacting with said system[12].

The research report was structured according to the recommendations of the Standards for Quality Improvement Reporting Excellence (SQUIRE 2.0), and the presentation of the data followed the quantitative and descriptive approach.

The study was conducted in the city of Campinas, São Paulo, Brazil, which has 68 UBSs (Health Centers) based on the Family Health Program (FHP) logic . These UBSs are distributed across six Health Districts, each responsible for health planning and management in areas with approximately

iversidade Estadual de Campinas
n of these units considered the link
d a Professional Doctorate Program
that guides the implementation of applied studies in the student's place of work.

This choice enabled the contextualized application of the object of study, considering the researcher's prior knowledge of the organizational structure, work processes and institutional flows involved in the topic investigated.

The ConectAPS-CRIE platform was developed to facilitate the management of requests for special immunobiologicals in PHC, replacing the need for these requests to be completed on paper or by email[15]. Its areas include Login, Home Panel, Users, Units, Immunobiologicals, Patients, Requests, Reports, and Help. Its use allows services to register requests, track their status, and attach files to requests, such as reports or prescriptions, vaccination records, and other digital documents. This tool was previously validated by IT professionals and nurses and improved based on nonconformities identified during the process[15]. However, as of the writing of this article, the platform had not yet been implemented in healthcare services.

The pilot phase began on April 10th and concluded on August 10th, 2024, and followed the municipality's existing request flow. The vaccination room professional forwards the request for the special immunobiological to the corresponding district VE. The VE is responsible for evaluating the request, which, if it meets the criteria for acceptance, is authorized and sent to the CRIE. After approval by the CRIE, the vaccine is transported to the CS. As soon as the vaccination room professional receives the special immunobiological, they contact the patient to arrange for its administration.

The role of each service in ordering and delivering special immunobiologicals was considered, involving the CS, the VE of the Northwest District, and the CRIE. The professionals of the services were trained by the researcher on the purpose and use of the platform, and the availability of an instruction manual and the possibility of contacting the researcher in case of doubts or difficulties were also discussed.

Data regarding the usability assessment were collected in the months of August and September 2024. Direct users of the software participated in the evaluation during the referred period. The pilot project consisted of a nurse, three nursing technicians, and a nursing assistant from the vaccination room at the CS, a nurse from the VE of the Northwest District, and two nurses from the CRIE. This constituted a

RESULTS

For a better understanding of the evaluated software, the main areas of the ConectAPS-CRIE platform (Home Panel and Requests) are presented in Figure 1.

cessing other areas, namely: Users,
ls, Patients, Requests, Reports, Help,

The Requests area contains a list of vaccines already requested for each patient, including the requesting unit, vaccine, date of birth, age, and request status. For each request, there are four icons in the Actions field: Status (view the request history), Attachments (view the file attached to the request or attach additional files), View (generates an individual report with patient and request data, which can be printed), Change (edit the request), and Delete. There is also the Insert New icon to add a new request.

During the pilot period, 11 patients and 14 requests for immunobiologicals were registered on the platform.

from CS, and one nursing assistant, also from CS.

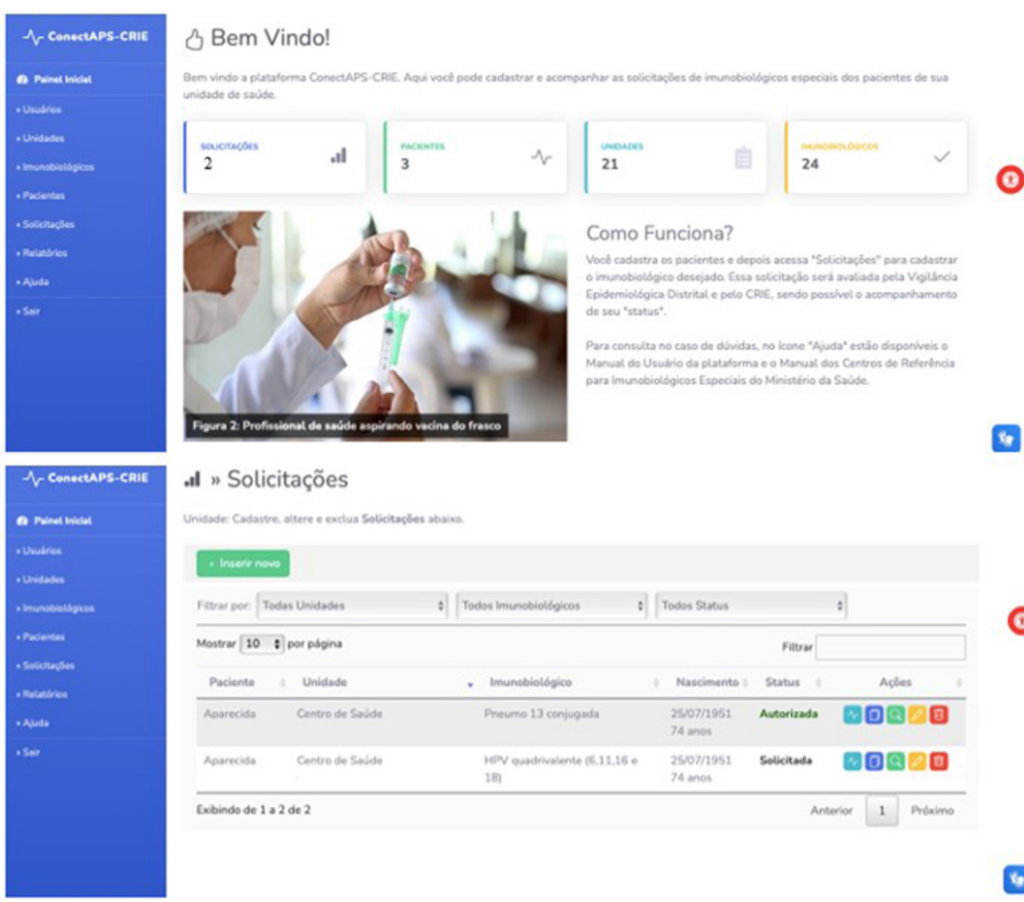
The distribution of responses given to the System Usability Scale statements by nursing professionals is represented in Figure 2.

The evaluation scores of the ConectAPS-CRIE platform by nursing professionals, using the System Usability Scale, are presented in Table 1.

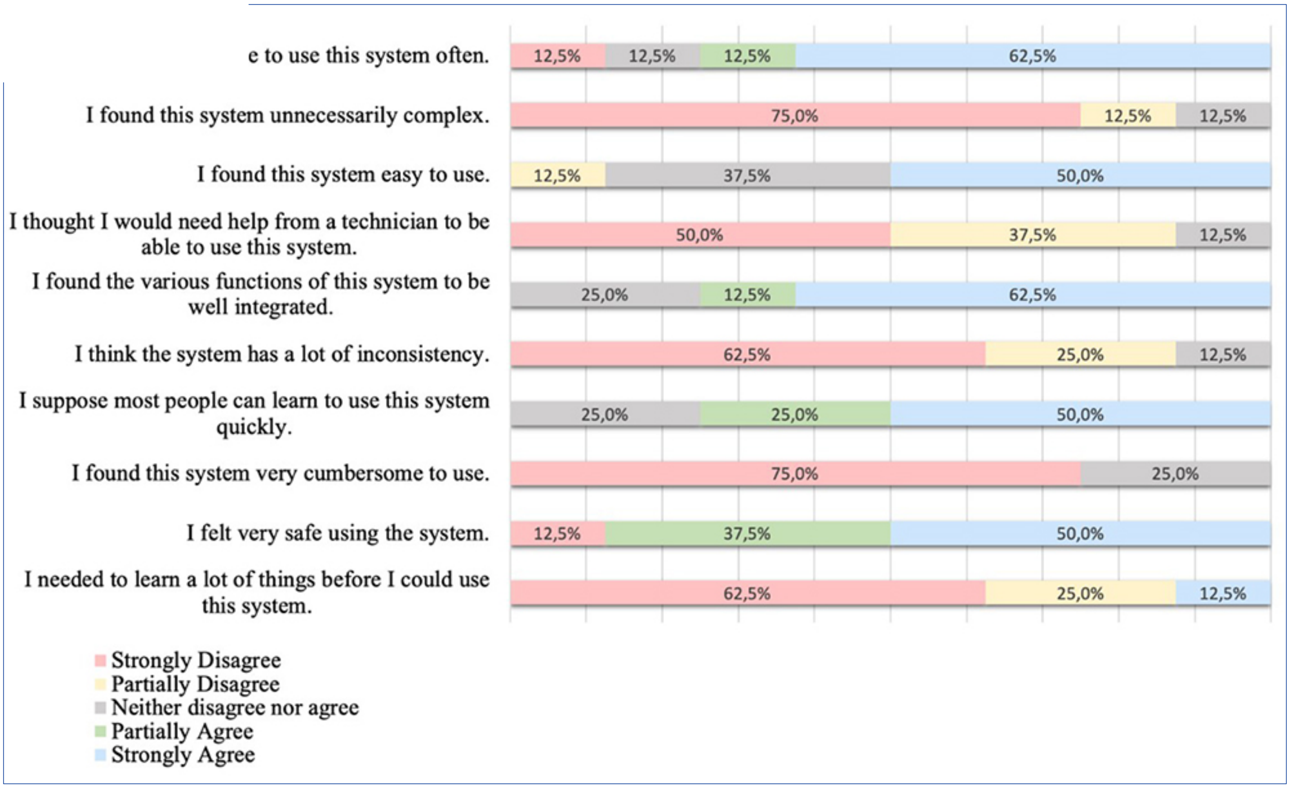
The classification and frequency of usability evaluation scores assigned to the ConectAPS-CRIE platform by nursing professionals are represented in Table 2.

The average score obtained was 82.5±17.9, allowing the usability assessment of ConectAPS-CRIE to be classified as excellent.

Figure 1. Illustration of the platform's Home Panel and Requests areas. Campinas, SP, Brazil, 2024



Source: authors' database, 2024



Source: Authors' database, 2024

Table 1. Evaluation scores of the ConectAPS-CRIE platform by nursing professionals (n=8), using the System Usability Scale. Campinas, São Paulo, Brazil, 2024

Professional	Workplace	Score (0-100)
Nurse 1	Epidemiological monitoring	77.5
Nurse 2	Health Center	97.5
Nurse 3	CRIE	47.5
Nurse 4	CRIE	67.5
Nursing technician 1	Health Center	97.5
Nursing technician 2	Health Center	100.0
Nursing technician 3	Health Center	87.5
Nursing Assistant 1	Health Center	85.0

Source: Authors' database, 2024

	N	%	Classification
	–	–	worst imaginable
26 - 39	–	–	poor
40 – 52	1	12.5	median
53 – 74	1	12.5	good
75 – 85	2	25.0	excellent
86 - 100	4	50.0	best imaginable

Source: Authors' database, 2024

DISCUSSION

This study evaluated the usability of the ConectAPS-CRIE platform from the perspective of the professionals involved in the services. It showed that nursing professionals (nurses, nursing technicians, and nursing assistants) assigned usability evaluation scores classified as average (12.5%), good (12.5%), excellent (25%), and best imaginable (50%). The average was 82.5±17.9 points, classified as excellent, indicating, in general, ease of use of the software. It is noteworthy that data collection, conducted immediately after the completion of the pilot phase, may have contributed to obtaining more accurate and reliable data, considering that participants' memories of the intervention remained fresh.

Participation in technology evaluation is essential, as it allows for the identification of weaknesses and guidance for improvements, making the software more appropriate for professional practice[17]. It is also important to ensure satisfactory levels of usability to enable its adoption in different care contexts, as the effective application of information and communication technologies in nursing practice still faces several challenges, such as difficulties in use, barriers to implementation and a lack of specialized technical support[18].

Similar technology evaluation studies were found in the literature that obtained even higher averages[13,19]. A study on the usability of an educational hypermedia on reception and obstetric risk classification, also using the System Usability Scale, obtained a higher average (91.9 points). However, the study had, as a limitation, the lack of participation of the target audience in the evaluation process[19].

ifficulties related to training or pre-
echnologies.

scores by location, it is observed that VE professionals assigned an intermediate value (77.5), while those from CS presented higher scores, ranging from 85 to 100, demonstrating greater familiarity or ease in using the platform, although with some internal variation. On the other hand, the two professionals linked to CRIE registered the lowest scores (47.5 and 67.5), which highlights a difference in the evaluation, depending on the activity carried out in each service.

High scores in the CS suggest a more favorable environment for using the platform, whether due to greater adherence, support, or professional profile. Lower scores in the CRIE may be related to the more specialized nature of the service, which requires, in addition to evaluating requests, managing the flow of information using the National Immunization Program Information System or a proprietary system with interoperability with the National Health Data Network (RNDS) to record the movement of special immunobiologicals[1].

It was observed that the highest usability scores were mostly assigned by mid-level and technical nursing professionals. However, the CS nurse had the highest score among the participating nurses. These data suggest that perception of usability is more related to the context of work than to educational level, contradicting the expectation that less academic training would result in greater difficulty using the software.

The importance of PHC nurses appropriating technological advances in service management is highlighted, as a way of qualifying assistance and improving the care provided to users[10]. Although still little explored, information technologies have been used by these professionals as organizational tools, with emphasis on the National Regulatory System (SISREG) and the Electronic Citizen Record (PEC)[10].

In this regard, the score assigned by the CS nurse suggests the acceptability of the platform, a consequence of its greater usability, in relation to other professionals. It is important to emphasize the importance of these professionals knowing, exploring and incorporating technologies into their practice to strengthen the organization of services[10].

A study conducted in India on the implementation of an integrated health information system for primary care highlighted initial difficulties in adapting to the digital system. Over time, however, professionals reported significant benefits, such as easier record searches, faster report generation, and decision-making support. The study reinforces the

From this perspective, the poor results obtained by CRIE professionals point to the need for further research into the factors that influence the platform's usability in this specific scenario. Thus, the study demonstrates that, for the implementation and adoption of new technological tools, studies are needed that explore their usability, considering the routines of healthcare professionals in different contexts.

In view of the results, it is clear that a limitation of the study was that the platform evaluation involved only one PHC unit and that its pilot was conducted over a short period of time. These factors limited the sample size, as a small number of professionals used the platform.

CONCLUSION

The usability assessment of the ConectAPS-CRIE platform, from the perspective of nursing professionals in the services involved, reveals, in general, good acceptance and ease of use, with an average score classified as excellent by the System Usability Scale. However, the variation in scores across different care settings suggests that the experience with the platform may be influenced by the profile of the professionals and the nature of the activities performed in each service.

Professionals working at the Health Center showed greater ease in using the tool, while those at the Reference Center for Special Immunobiologicals assigned the lowest scores, highlighting specific challenges related to the complexity of this service.

The findings reinforce the need for further research that explores, in a broader and more representative manner, the factors that affect the usability of the ConectAPS-CRIE platform in different contexts within the healthcare network. It is important to consider further adjustments to ensure its widespread acceptance and use. Such improvements will be essential for the platform to establish itself as an effective solution for managing these immunobiologicals, aiming to meet the specific demands of each service.

After improvements are made, the platform's implementation is planned to be expanded across several PHC units and specialized services, aiming to increase usage and user numbers, and to understand the specific nature of requests for special immunobiologicals. This way, the platform can improve workflows and make request management more accessible to services, making it easier for the professionals involved, and potentially impacting increased vaccination coverage among populations with special clinical conditions.

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The authors declare that there is no conflict of interest.

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